

Assignment Sheet

(Session: 2023-24)

Assignment No	string	Subject	string
Branch	string	Semester	string
Unit	Multiple Units	Faculty	string
Date Given	string	Submission Date	string

Q.No	Questions	CO	Marks
Q1	Explain the process of data normalization in time series analysis and its significance in model training. Provide an example of how mean absolute error (MAE) is used to evaluate the accuracy of time series forecasts	CO1	5
Q2	Discuss the differences between ARIMA and exponential smoothing models in handling time series data. How do these models account for seasonality and trends in the data?	CO2	5
Q3	Justify the use of Long Short-Term Memory (LSTM) networks for sequence modeling in natural language processing tasks. Compare their performance with traditional RNNs in capturing long-term dependencies	CO1	5
Q4	Describe the process of feature engineering in the context of making predictions using linear regression. Provide an example of a scenario where decision trees might be more appropriate than linear regression	CO2	5
Q5	Evaluate the pros and cons of using neural networks for making predictions in highly imbalanced classification problems in time series data	CO1	5

Department of Artificial Intelligence

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